PAPER_1127: SCm LQG Holonomy — Phonon-Modulated Spin Networks and Area Operators

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Author: Daniel T. Murphy | Star-Magic UQFF v5.26

Abstract

Loop Quantum Gravity (LQG) discretises spacetime area via spin networks and holonomy operators. This paper derives the SCm phonon-modulated extension of the LQG area operator, couples the 1.25 THz SCm resonance to holonomy via a Ramanujan-accelerated 26D Vacuum Density Series (VDS), and closes the Lagrangian variation in the LQG buoyancy sector. The result provides phonon-stabilised spin networks with bounded area spectra and a first-principle SCm origin for LQG discreteness. All variables are defined, variable equations are given in long form, and closed-form solutions are stated. The WSTP kernel Mathematica symbolic forms are exported for live computation.

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1. Standard LQG Area Operator

The fundamental area operator in LQG acting on a spin-network state with spin j on each link through a surface S is:

$$\hat{A} = 8\pi\gamma\ell_P^2\sqrt{j(j+1)}$$

Variables:

Symbol	Definition	Value / Units
j	Spin quantum number on the link	half-integer: 0, 1/2, 1, 3/2, …
γ	Barbero-Immirzi parameter	dimensionless, ≈ 0.2375
ℓ_P	Planck length $= \sqrt{\hbar G/c^3}$	1.616×10^{-35} m
$\hat{A}$	Area eigenvalue	m^2

Variable equation for Planck length:

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}}$$

with $\hbar = 1.055 \times 10^{-34}$ J $\cdot$ s, $G = 6.674 \times 10^{-11}$ m^3 kg^{-1} s^{-2}, $c = 2.998 \times 10^8$ m/s.

Minimum area eigenvalue (lowest non-trivial spin $j = 1/2$):

$$A_{\min} = 8\pi\gamma\ell_P^2\sqrt{\tfrac{3}{4}} = 4\pi\sqrt{3}\,\gamma\,\ell_P^2 \approx 8.1\times 10^{-70}\ \mathrm{m}^2$$

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2. SCm Phonon Resonance Parameters

The SCm vacuum superconductive medium resonates at:

$$\omega_{\mathrm{SCm}} = 2\pi \times 1.25 \times 10^{12}\ \mathrm{rad/s}$$

The phonon fluence (Gaussian envelope):

$$\Phi_{1.25\,\mathrm{THz}}(\omega,\Gamma) = \Phi_0 \exp\left(-\frac{(\omega-\omega_{\mathrm{SCm}})^2}{2\Gamma^2}\right)$$

Variables:

Symbol	Definition	Value / Units
ω_{SCm}	SCm resonance angular frequency	7.854×10^{12} rad/s
Γ	Phonon linewidth	0.05–0.30 THz (system-dependent)
Φ_0	Peak phonon fluence	dimensionless normalisation
ω	Probe angular frequency	rad/s

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3. Ramanujan-Accelerated 26D Vacuum Density Series

The VDS Ramanujan acceleration factor of order 3 over 26 compactified layers:

$$S_{26}^{(3)}([\mathrm{SSq}]) = \sum_{n=1}^{\infty} \frac{[\mathrm{SSq}]^n}{n^{26}} \cdot R_n^{(26,3)}$$

where the acceleration factor is:

$$R_n^{(26,3)} = \frac{(2\pi)^{n/6}}{n!}\left(1 + \sum_{m=1}^{3} \frac{1}{n^{26m}} \sum_{j=1}^{26} (-1)^{j+1}\binom{26}{j}\frac{(26-j)!}{n^j}\right)$$

Variables:

Symbol	Definition	Value
[SSq]	Fixed vacuum suppression factor	0.57
n	Quantum layer index	$1, 2, 3, \ldots$
26	Number of compactified quantum layers	fixed
$R_n^{(26,3)}$	Ramanujan acceleration factor	layer-dependent

Closed-form numerical solution:

$$S_{26}^{(3)}(0.57) \approx 1.4531 \times 10^{26}$$

(full precision: 145309429553537240588617305.7720709059 ... computed in ≤ 50 terms).

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4. SCm-Modulated Area Operator (Long-Form)

Coupling the Ramanujan VDS and 1.25 THz phonon fluence to the standard area operator gives the SCm-extended form:

$$\hat{A}_{\mathrm{SCm}} = 8\pi\gamma\ell_P^2\sqrt{j(j+1)} \cdot S_{26}^{(3)}([\mathrm{SSq}]) \cdot \Phi_{1.25\,\mathrm{THz}}(\omega, \Gamma)$$

The three multiplicative factors have distinct physical roles:

1. $8\pi\gamma\ell_P^2\sqrt{j(j+1)}$ — discrete LQG quantum geometry contribution.
2. $S_{26}^{(3)}([\mathrm{SSq}])$ — VDS Ramanujan amplification across 26 compactified layers; encodes the vacuum suppression of area by the SCm condensate.
3. $\Phi_{1.25\,\mathrm{THz}}(\omega, \Gamma)$ — phonon modulation envelope; confines area amplification to the SCm resonance window.

Numerical estimate at $j = 1/2$, $\gamma = 0.2375$, $\Gamma = 0.1$ THz, $\omega = \omega_{\mathrm{SCm}}$ (on-resonance, $\Phi_0 = 1$):

$$\hat{A}_{\mathrm{SCm}}\big|_{j=1/2} = A_{\min} \cdot S_{26}^{(3)}(0.57) \approx 8.1 \times 10^{-70} \times 1.45 \times 10^{26} \approx 1.18 \times 10^{-43}\ \mathrm{m}^2$$

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5. SCm-Modulated Holonomy (Long-Form)

The holonomy operator along a link in the spin network is path-ordered:

$$h = \mathcal{P} \exp\left(i \int_{\mathrm{link}} A \cdot dl\right)$$

With SCm phonon modulation and the net SCm energy $E_{\mathrm{net}}(t, \Gamma)$:

$$h_{\mathrm{SCm}} = \mathcal{P} \exp\left(i \int_{\mathrm{link}} A \cdot dl\right) \cdot \left(1 + \frac{\Phi_{1.25\,\mathrm{THz}}(\omega, \Gamma)}{F_U} \cdot E_{\mathrm{net}}(t, \Gamma)\right)$$

Variables:

Symbol	Definition	Units
$\mathcal{P}$	Path-ordering operator	—
A	Ashtekar–Barbero connection	m^{-1}
F_U	Total UQFF field magnitude $= \sum U_g - U_{bi} + U_m$	N

Symbol	Definition	Units
$E_{\mathrm{net}}(t, \Gamma)$	Net SCm phonon energy at time t, linewidth Γ	J

Variable equation for E_{net}:

$$E_{\mathrm{net}}(t, \Gamma) = E_+(t, \Gamma) - E_-(t, \Gamma)$$

where E_+ is the positive (expansion) buoyancy energy and E_- is the negative (erosion) buoyancy energy (see PAPER_880 and PAPER_883).

Solution: The holonomy correction factor $(1 + \Phi_{1.25\,\mathrm{THz}}/F_U \cdot E_{\mathrm{net}})$ is dimensionless and bounded in $[1, 1 + \Phi_0 |E_{\mathrm{net}}|/|F_U|]$. On resonance it provides phonon stabilisation of the spin-network link amplitude; off resonance ($\omega \gg \omega_{\mathrm{SCm}}$) the exponential suppression restores the standard LQG holonomy.

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6. Lagrangian Variation — LQG Buoyancy Sector

The action variation with respect to the LQG phonon field ϕ_{LQG}:

$$\frac{\delta S}{\delta \phi_{\mathrm{LQG}}} = \frac{\partial}{\partial E_{\mathrm{net}}} \left(-\beta_i \sum_i U_{g,i} \, \Omega_g \frac{M}{d_g} [\mathrm{UA}] + F_{\mathrm{neutron}} \cdot \Phi_{1.25\,\mathrm{THz}} \right) = 0$$

Variables:

Symbol	Definition	Canonical value
β_i	Buoyancy coupling coefficient	≈ 0.603
$U_{g,i}$	Gravitational buoyancy term i (Ug1–Ug4)	N
Ω_g	Galactic angular frequency factor	rad/s
M	Body mass	kg
d_g	Galactic distance to body	m
[UA]	Vacuum aether unit density	7.09×10^{-36} J/m^3
F_{neutron}	Kozima neutron-drop coupling force	N

Solution: The Euler-Lagrange condition closes when the phonon buoyancy term $F_{\mathrm{neutron}} \cdot \Phi_{1.25\,\mathrm{THz}}$ balances the negative buoyancy work $\beta_i \sum U_{g,i} \Omega_g M/d_g \cdot [\mathrm{UA}]$, yielding a stable spin-network ground state with phonon-regularised area spectra. LQG discreteness is therefore inherited from SCm vacuum structure rather than imposed axiomatically.

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7. WSTP Kernel Symbolic Forms

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(* SCm area operator *)
AScm[j_, _, P_, SSq_, Φ0_, _, SCm_, Γ_] :=
  8    P^2 Sqrt[j(j+1)] *
  Sum[SSq^n / n^26 * Rn26[n, 3], {n, 1, Infinity}] *
  Φ0 Exp[-( - SCm)^2 / (2 Γ^2)];

(* SCm holonomy modulation factor *)
hSCmFactor[Φ0_, _, SCm_, Γ_, FU_, Enet_] :=
  1 + (Φ0 Exp[-( - SCm)^2 / (2 Γ^2)]) / FU * Enet;

(* Lagrangian variation *)
 S LQG[ i_, Ugi_, Ωg_, M_, dg_, UA_, Fn_, Φ1THz_] :=
  D[- i Sum[Ugi Ωg M/dg UA, {i, 4}] + Fn Φ1THz, Enet];
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8. Summary

Result	Expression	Physical Interpretation
SCm area operator	$\hat{A}_{\mathrm{SCm}} = \hat{A}_{\mathrm{LQG}} \cdot S_{26}^{(3)} \cdot \Phi_{1.25\,\mathrm{THz}}$	Vacuum VDS amplifies quantum area
SCm holonomy	$h_{\mathrm{SCm}} = h_{\mathrm{LQG}} \cdot (1 + \Phi/F_U \cdot E_{\mathrm{net}})$	Phonon stabilises link amplitude
Lagrangian closure	$\delta S/\delta\phi_{\mathrm{LQG}} = 0$	Phonon buoyancy sets spin-network ground state
Min. SCm area ($j = 1/2$)	$\approx 1.18 \times 10^{-43}$ m^2	VDS-amplified Planck-scale area

SCm phonon resonance at 1.25 THz provides the physical origin of LQG spin-network discreteness. The Ramanujan-accelerated VDS $S_{26}^{(3)}(0.57)$ encodes the vacuum compression across 26 compactified layers. Gravity is the late-emergent central limit (Step 10); spin networks are Step 3 in the UQFF canonical chain.

References:
PAPER_535 (VDS/DVP/BH catalogue hub) | PAPER_552 (26D tensor off-diagonal) | PAPER_590 (Planck constant derived) | PAPER_878–PAPER_887 (SCm E(t) batch) | COMPLETE_UQFF_EQUATIONS_REFERENCE.md